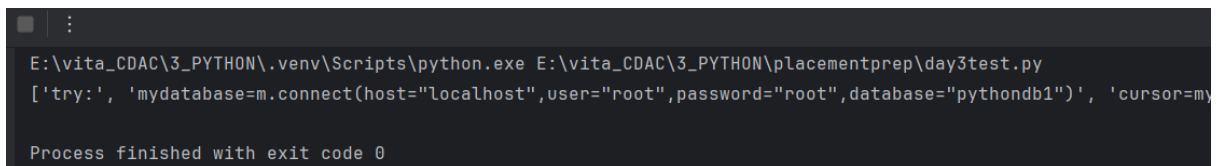


Q.1) Write a code to Read a file and append lines to a list.

```
lines_list = []
with open("try.txt", "r") as file: # Replace 'sample.txt' with your filename
    for line in file:
        lines_list.append(line.strip()) # strip() removes newline characters

print(lines_list)
```



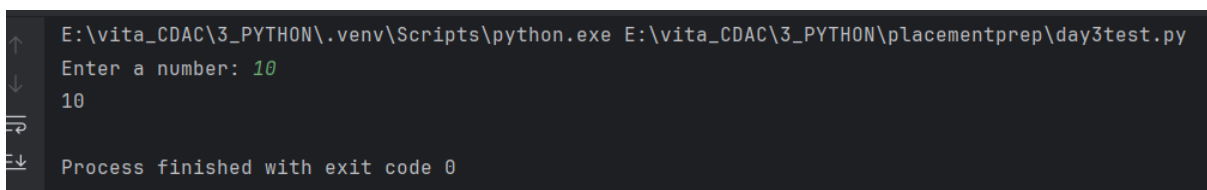
```
E:\vita_CDAC\3_PYTHON\venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
['try:', 'mydatabase=m.connect(host="localhost",user="root",password="root",database="pythondb1")', 'cursor=my']

Process finished with exit code 0
```

Q.2) Write a code to catch an Exception in python?

```
from exceptiongroup import catch

try:
    num = int(input("Enter a number: "))
    print(num)
except ValueError as e:
    print("invlaid input")
```



```
E:\vita_CDAC\3_PYTHON\venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
Enter a number: 10
10

Process finished with exit code 0
```

Q.3) Write a Python function that accepts a list containing strings and integers.

Merge all string elements using # and add all integer elements.

e.g.

input list is

```
['100', 'welcome', 'hi', '200', '300', 'bye', 'welldone', '500']
```

Output should be:

```
welcome#hi#bye#welldone#
```

```
1100
```

```
def process_list(input_list):
    strings = []
    total = 0

    for item in input_list:
        if item.isdigit(): # Check if the element is a number (string form)
            total += int(item)
        else:
            strings.append(item)

    merged_strings = "#".join(strings) + "#" # Add trailing #
    return merged_strings, total

# Example usage
input_list = ['100', 'welcome', 'hi', '200', '300', 'bye', 'welldone', '500']
result = process_list(input_list)
print(result[0]) # Output: welcome#hi#bye#welldone#
print(result[1]) # Output: 1100
```

```
E:\vita_CDAC\3_PYTHON\.venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
welcome#hi#bye#welldone#
1100

Process finished with exit code 0
```

Q.4) Write a script to sort a dictionary based on its values and find the sum of middle two values

```
input_dict = {"x": 5, "y": 15, "z": 25}
```

Output:

```
Sorted Dictionary: {'x': 5, 'y': 15, 'z': 25}
```

```
Sum of middle two values: 15 + 5 = 20
```

or

```
input_dict = {"x": 5, "y": 15, "z": 25, "p": 12}
```

Output:

```
Sorted Dictionary: {'x': 5, 'p': 12, 'y': 15, 'z': 25}
```

```
Sum of middle two values: 12 + 15 = 27
```

```
def sort_dict_and_sum_middle(input_dict):
    # Sort dictionary by values
    sorted_dict = dict(sorted(input_dict.items(), key=lambda x: x[1]))
```

```

print("Sorted Dictionary:", sorted_dict)

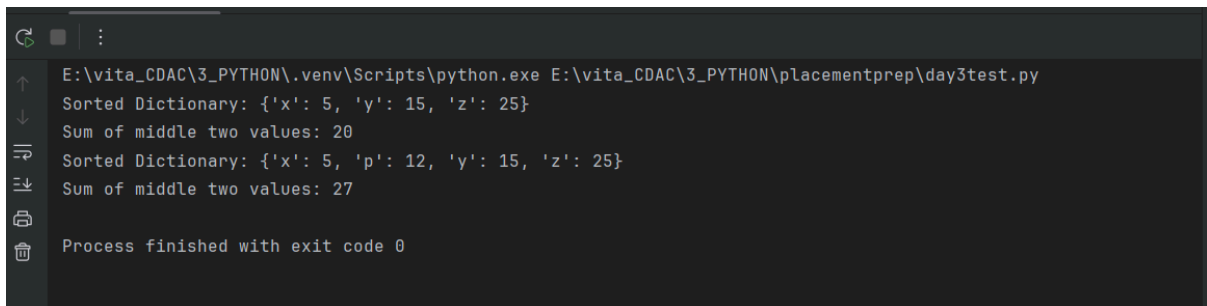
# Extract values
values = list(sorted_dict.values())

# Find middle two values
n = len(values)
if n % 2 == 0: # Even number of elements
    middle_sum = values[n // 2 - 1] + values[n // 2]
else: # Odd number of elements
    middle_sum = values[n // 2 - 1] + values[n // 2] # As per given examples

print("Sum of middle two values:", middle_sum)

# Example usage
sort_dict_and_sum_middle({"x": 5, "y": 15, "z": 25})
sort_dict_and_sum_middle({"x": 5, "y": 15, "z": 25, "p": 12})

```



```

E:\vita_CDAC\3_PYTHON\.venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
Sorted Dictionary: {'x': 5, 'y': 15, 'z': 25}
Sum of middle two values: 20
Sorted Dictionary: {'x': 5, 'p': 12, 'y': 15, 'z': 25}
Sum of middle two values: 27
Process finished with exit code 0

```